

Experiment 04 — ExpressJS, Mongoose, CRUD and REST API

GitHub-readable preview of the presentation. Download the matching .pptx for the editable PowerPoint.

1. Syllabus tasks

- Connect to MongoDB.
- Define a Product schema.
- Create products.
- Read all products.
- Update a product.
- Delete a product.
- Display the API data in a simple browser page using fetch().

2. Workflow

Experiment 04 — ExpressJS, Mongoose, CRUD and REST API



Workflow: input → processing → data/state → output

3.1. Code pathway — server.js

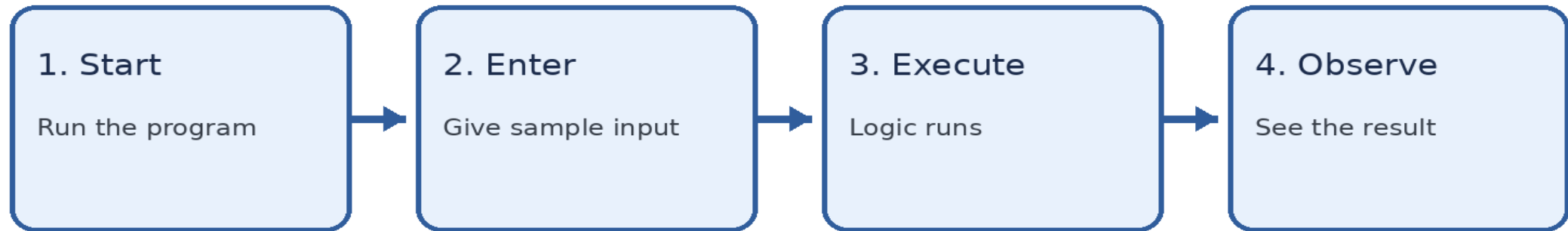
```
1 /**
2  * Experiment 04: ExpressJS + Mongoose + CRUD + REST API
3  *
4  * What it demonstrates:
5  * - MongoDB connection using Mongoose
6  * - Mongoose schema and model
7  * - GET, POST, PUT and DELETE REST endpoints
8  * - A simple browser page that consumes the REST API
9  */
10
```

3.2. Code pathway — server.js

```
11 const express = require('express');
12 const mongoose = require('mongoose');
13
14 const app = express();
15
16 const PORT = 3004;
17 const MONGO_URI =
18 process.env.MONGO_URI ||
19 'mongodb://127.0.0.1:27017/fsd2lab';
20
```

4. Runtime walkthrough

Runtime walkthrough — Experiment 04 — ExpressJS, Mongoose, CRUD and F



Runtime: start → enter → execute → observe

5. Run and observe

- Start the server / Vite app or open the MongoDB shell as described in the experiment README.
- Enter the sample input or call the required route.
- Watch the terminal for logs and errors.
- Verify the browser, API response, or database result.

6. Viva questions

- What is the purpose of this experiment?
- Which input is required?
- Which line or command performs the main operation?
- What output should you observe?
- What happens when the input changes?